UNIVERSITY OF LIVERPOOL
DEPARTMENT OF ELECTRICAL ENGINEERING & ELECTRONICS

Project Plan for project 'Fault-Tolerant Control of a Reaction Wheel System for CubeSat Attitude Stabilisation'

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Tables of contents, figures, etc.

Glossary

Term	Definition
Reaction Wheel Array (RWA)	A group of two or more reaction wheels working together to control an attitude or orientation
Hardware In Loop (HIL)	A type of testing that involves running a simulation with a hardware component alongside the simulated model
Attitude Determination and Control System (ADCS)	A system that determines the orientation and points the system in a specific direction
Computer Aided Design (CAD)	Computer software used in the modelling and design of mechanical components
Degrees of Freedom (DOF)	The number of independent movements available
Inertial Measurement Unit (IMU)	A system used to determine position and orientation in free space
Micro Controller Unit (MCU)	A self-contained computer used for embedded applications
Root Mean Squared (RMS)	Calculation to find the magnitude of a varying signal
Fault Detection and Isolation (FDI)	Detecting abnormal or unexpected behaviour in a system and acting to mitigate it
Pulse Width Modulation (PWM)	A digital control technique that delivers a desired power through varying duty cycle

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Project scope

Project Aim and Goals:

This project aims to deliver the design, simulation, fabrication and evaluation of a satellite Attitude Determination and Control System (ADCS) module. Specifically, it aims to deliver an over-actuated reaction wheel system, allowing for fault-tolerant control of a small CubeSat. It will achieve this through the use of reaction wheels. These are motor driven flywheels that generate angular momentum in order to rotate an object through conservation of momentum [1]. The project should be able to demonstrate attitude control along a commanded axis in the event of failure in one reaction wheel or degradation of performance.

Current State of the Art

In recent years, small-scale, mass deployable satellites known as "CubeSats" have become a significant platform for space payloads among academia and the commercial sector. CubeSats are a common platform standard for micro-satellites, offering a volume of 10cm cubed for a single CubeSat unit (1U). Due to their low weight and volume, CubeSats offer a significantly cheaper platform for space applications, making them increasingly popular over recent years. Over 2730 CubeSats have been launched into orbit since 2003 [2].

Advancements in capability of CubeSats have been possible through newer and more advanced commercial components, and as a result, the demand for CubeSat ADCS modules is growing. These systems allow satellites to perform re-orientation and tracking for sensing and imaging tasks. The most common actuators found in micro-satellites are reaction wheel arrays (RWA), with over-actuated systems becoming more popular. Over-actuated reaction wheel systems are sought after due to their fault-tolerance, and the increased risk of failure presented by mechanical wear in reaction wheels. These over-actuated systems have a surplus of actuators (in this case reaction wheels) for the three axes of rotation a spacecraft has. This allows the actuators to act on all three planes of rotation, even if one actuator fails, maintaining full control of the satellites attitude.

Other actuation methods exist, most notably, magnetorquers, which are wire or PCB trace coils used to generate electromagnetic fields to act against the magnetic field of a nearby body. These offer a coarser control method compared to reaction wheels, but are less prone to failure and unlike reaction wheels, are able to desaturate angular momentum [3]. Other actuators like control moment gyros and thrusters have their own advantages or disadvantages, but are usually too large to implement on CubeSats. This project will only focus on reaction wheels and torque reconfiguration of a RWA for fault-tolerant control.

Currently, many over-actuated systems that are available commercially are high-precision, custom-built modules. They offer excellent performance but may be financially infeasible to small teams building their own CubeSat subsystems. Furthermore, their proprietary nature makes the technology less accessible to smaller academic institutions and CubeSat teams. This project aims to address this by achieving the design of advanced ADCS systems without the use of bespoke hardware, reducing the entry barriers for future micro-satellite missions.

Considerations

CubeSats and their subsystems must address many of the wider issues often faced by spacecraft and their missions. A significant challenge is mitigating the environmental impact of such systems. Artificial satellites in large numbers can have a negative ecological impact. The fabrication and launch of satellites emit greenhouse gasses and eject particulate matter into the atmosphere [4]. Increasing numbers of satellites in orbit also raise concerns on their potential impact on ground-based astronomy, as satellites reflect light brightly. This may disrupt scientific research in the future as satellite constellations and debris increase [5].

Fault-tolerant ADCS hardware may improve the lifecycle of micro-satellites by enabling them to operate for longer lifespans, and enable manoeuvres for deorbiting, thereby reducing excess debris through removal of material and improved environmental impact via reduction of satellite launches. An example of a reaction wheel failure limiting the mission capabilities of a spacecraft was on the Keplar spacecraft [6], which lost its mission capability after losing two of its four reaction wheels. However, before the failure of a second wheel, the RWA allowed the spacecraft to continue its mission, demonstrating the value of redundant reaction wheel design.

Project objectives

Overall

To design, simulate, fabricate and test a fault-tolerant reaction wheel system for CubeSat attitude stabilisation, using reconfigurable torque allocation to maintain a commanded attitude within 5° in under 30 seconds during a simulated failure of a reaction wheel without overshoot exceeding 20°, leveraging off-the-shelf components, thereby enhancing the reliability and operational lifespan of low-cost micro-satellites, by March 2025.

Reaction Wheel Design

Design a reaction wheel that can be placed in a tetrahedral (pyramid) configuration comprised of four reaction wheels total in a volume less than 100mm x 100mm x 150mm. Select a motor and design a flywheel that can reliably spin over or equal to 6000 rpm to produce a nominal angular momentum greater than or equal to $0.003 Kgm^2s^{-1}$ [7]. As well as a peak momentum of at least $0.005 Kgm^2s^{-1}$ based on commercial reaction wheels used for similar roles [7]. By November 2025.

System Fault-Tolerance

Design and test a control system capable of detecting and isolating faults in the reaction wheels. It should reconfigure torque allocation, enabling the system to maintain full attitude control for a catastrophic failure of one reaction wheel. Being either non-operational, or stuck rotating at full speed. It should also compensate for degraded performance in reaction wheels,

being able to maintain a commanded attitude within 5° with a total degradation of 30% across the RWA. By February 2026.

System Modelling and Simulation Framework

Develop a 6-DOF simulation of CubeSat dynamics with realistic wheel inertia and fault scenarios to evaluate control system performance. Validate the model against data from HIL testing, ensuring the simulated attitude error remains within 10% of experimental results. By February 2025.

Cost

In total the overall cost of the project should not exceed £1000, as it is meant to demonstrate fault-tolerant ADCS subsystems for space missions performed by academic institutions and small organisations. This enables micro-satellite missions to operate for an extended lifespan, which gives these organisations a larger return on their initial investment into such missions. This criterion is to be upheld throughout the entire project, until March 2026.

Project Requirements

Category Reaction Wheel Performance Requirements	
Identifier	**Description**
RW-01	Speed range 0-6000rpm or greater
RW-02	Able to generate a nominal angular momentum of 0.003Kgm^{2}s^{-1} or greater
RW-03	Total dimensions in tetrahedral configuration < 150mm x 100mm x 100mm (All four reaction wheels)
RW-04	Produce a peak momentum of at least 0.005Kgm^{2}s^{-1}
Angle and Wheel Sensing Requirements	
SEN-01	IMU Measurement ranges up to Gyro ±500 °/s, Accel ±8 g, Sample rate >= 100Hz, Gyro noise <= 0.05 °/s/√Hz, bias drift <= 10 °/hr
SEN-02	IMU 9-DOF Accelerometer, Gyroscope, Magnetometer
SEN-03	Hall sensors provide raw counts to MCU; processed to compute flywheel angular velocity.
SEN-04	IMU filter produces quaternions to remove gimbal lock
Control System Performance	
CON-01	Demonstrate reconfigurable torque allocation after a wheel has been declared unhealthy, maintaining attitude error within limits
CON-02	Fault Detection and Isolation: Monitoring wheel speed and use linear approximation to detect faults in the performance of reaction wheels, detection time <= 2s
CON-03	Attitude error <= 5° with a total RWA degradation of 30% or complete failure of one reaction wheel
CON-04	Steady state pointing error <= 0.5° RMS
CON-05	Commanded slew of 10° completed settling within error band with a settling time <= 5s
Simulation	
SIM-01	6-DOF rigid body simulation of CubeSat model and reaction wheels dynamics
SIM-02	HIL integration of IMU, controllers and motors to be validated with real-time plant emulation

SIM-03	Various fault scenarios for reaction wheels, including; stalled wheel, stuck throttle and degraded performance simulations

WORK PACKAGE DESCRIPTION			
WP Title:	Mechanical Design and Electronics	**WP number**	1
Start Event:	14/10/2025		
Duration:	3 Weeks		

Objectives:

Design the mechanical components of a CubeSat reaction wheel module. Design and implement a sensing circuit to measure wheel rotational speed and attitude/angle of device. Implement a sufficient motor and motor controller for each reaction wheel.

Task Description:

- Produce a CAD model of the reaction wheel module
- Design the module structure to fit inside a 3U CubeSat (Maximum volume 150mm x 100mm x 100mm)
- Fabrication of mechanical design (3D Printing, machining of wheels)
- Design a circuit for sensor feedback (attitude, wheel rpm) and motor control
- Selection of electronic components
- Ordering of electronic components
- Breadboard testing
- PCB design and manufacture

Inputs:

- IMU Sensor Output Parameters (Datasheet)
- Hall Sensor Output Parameters (Datasheet)
- Motor/ Motor Controller Input Parameters (Datasheet)
- Kinematic requirements

Interfaces/links with other tasks or WPs:

WP3

Deliverables:

- 3D models and design drawings of reaction wheel module
- Fabricated components and assembly of mechanical design (prototype)
- Circuit diagram
- Protype PCB

Work packages

WORK PACKAGE DESCRIPTION			
WP Title:	Sensor and Actuator Interface	**WP number**	2
Start Event:	Completion or near completion of WP1		
Duration:	1-2 Weeks		

Objectives:

Stream IMU and Hall sensor data to the microcontroller for filtering and processing. The IMU data filtered to produce quaternions for positioning, and the hall sensor counts processed into angular velocities for each reaction wheel. Interface the reaction wheel motors through an appropriate motor driver to control their speed and direction through PWM signals.

Task Description:

- Connect to and receive data from the IMU
- Filter IMU data and convert to quaternions
- Connect to and receive data from the hall sensors
- Convert hall sensor data into angular velocities for each flywheel

Inputs:

- IMU minimum characteristics from project objectives
- Output parameters of IMU (for filtering)

Interfaces/links with other tasks or WPs:

WP1

Deliverables:

- Attitude data in quaternions
- Flywheel velocity data

WORK PACKAGE DESCRIPTION			
WP Title:	Dynamics Modelling and Control Design	**WP number**	3
Start Event:	14/10/2025		
Duration:	3-4 Weeks		

Objectives:

Develop kinematic and dynamic models to simulate the behaviour of a small satellite and its reaction wheels in Simulink/MATLAB. Design a control system for momentum allocation and tune values for optimised control. Choose and design a non-linear controller for the control loop.

Task Description:

- Calculate reaction wheel requirements for 3kg 3U CubeSat
- Determine kinematics and dynamics of satellite model
- MATLAB/Simulink model of satellite and reaction wheels kinematics + dynamics
- Implement a cascaded control system design
- Implement a non-linear controller
- Tune controller for optimal control for wheel configurations (All four and one failure mode)

Inputs:

- Reaction wheel performance/dimensions
- CubeSat physical parameters

Interfaces/links with other tasks or WPs:

None/Independent

Deliverables:

- Kinematics and dynamics formulae
- MATLAB code modelling the satellite design
- Simulink model of controller
- Simulations of motion and nominal controller performance

WORK PACKAGE DESCRIPTION			
WP Title:	Fault Detection and Isolation and Torque Allocator	**WP number**	4
Start Event:	Immediately after WP3		
Duration:	2-3 Weeks		

Objectives:

Design and implement a torque allocator for the RWA control system, choose a method for reconfiguration (e.g. Pseudo-inverse or quadratic programming). Program a fault detection and isolation programme that evaluates wheel health based on throttle input and measured velocity.

Task Description:

- Choose an allocation method for torque reconfiguration
- Design and Implement torque allocator
- Design and implement a fault detection and isolation programme
- Implement a script to log reaction wheel performance on calibration for FDI logic
- Perform limited testing to confirm functionality of control system and FDI code

Inputs:

- Reaction Wheel metrics
- Hall sensor data

Interfaces/links with other tasks or WPs:

WP3

Deliverables:

- Design document of control system overview, allocation algorithm choice, formulae
- FDI code and unit tests
- Calibration and logging script

WORK PACKAGE DESCRIPTION			
WP Title:	Simulation and Fault Injection	**WP number**	4
Start Event:	Immediately after WP4		
Duration:	2-3 Weeks		

Objectives:

Simulate different failures and degradation for a single wheel for different torque allocations. Simulate different scenarios (e.g. initial rotational velocity of 0.01 rad/s, wheel stuck at full throttle… etc.). Evaluate performance of the system in response to each scenario.

Task Description:

- Simulate the physics of the CubeSat model with the reaction wheels
- Run various simulations of fault scenarios, testing performance in the event of a wheel stall, throttle lock-up or degradation
- Log data on response times, control system performance, attitude errors… etc
- Perform unit testing for each software component

Inputs:

- Reaction wheel models and parameters
- Control system modules, FDI logic

Interfaces/links with other tasks or WPs:

WP3, WP4

Deliverables:

- Simulation suite and unit testing
- Performance report
- Demonstration

WORK PACKAGE DESCRIPTION			
WP Title:	Testbed Fabrication and HIL Evaluation	**WP number**	6
Start Event:	27/01/2026		
Duration:	2-3 Weeks		

Objectives:

Construct a test platform for the hardware prototype of the reaction wheel module. This can be either a single axis (suspended by string/ chain) or 3-axis (full gimbal) depending on time and material limitations. Evaluate the performance of the hardware by using the testbed and comparing the response measured with the simulated model.

Task Description:

- Mounting rig design to interface the plate of the RWA with the testbed
- Testbed design (Single axis option): design a mount for suspended or pivot-mounted frame
- Testbed design (3-axis option): design a 3-axis gimbal with mechanical bearing for reduced friction
- Fabricate and assemble testbed
- Integrate electronics with MATLAB/Simulink for HIL testing
- Perform calibration of IMU and reaction wheel performance
- Perform hardware testing of system performance in both nominal operation and during previous fault scenarios
- Analysis of performance and report

Inputs:

- Mechanical and electrical components
- Simulation framework
- Control system, calibration scripts and FDI logic

Interfaces/links with other tasks or WPs:

WP1, WP2, WP3, WP4, WP5

Deliverables:

- Physical testbed + CAD models
- HIL interface
- Demo data logs
- Performance evaluation report
- Photo or video of operation

Skills and Resource Audit

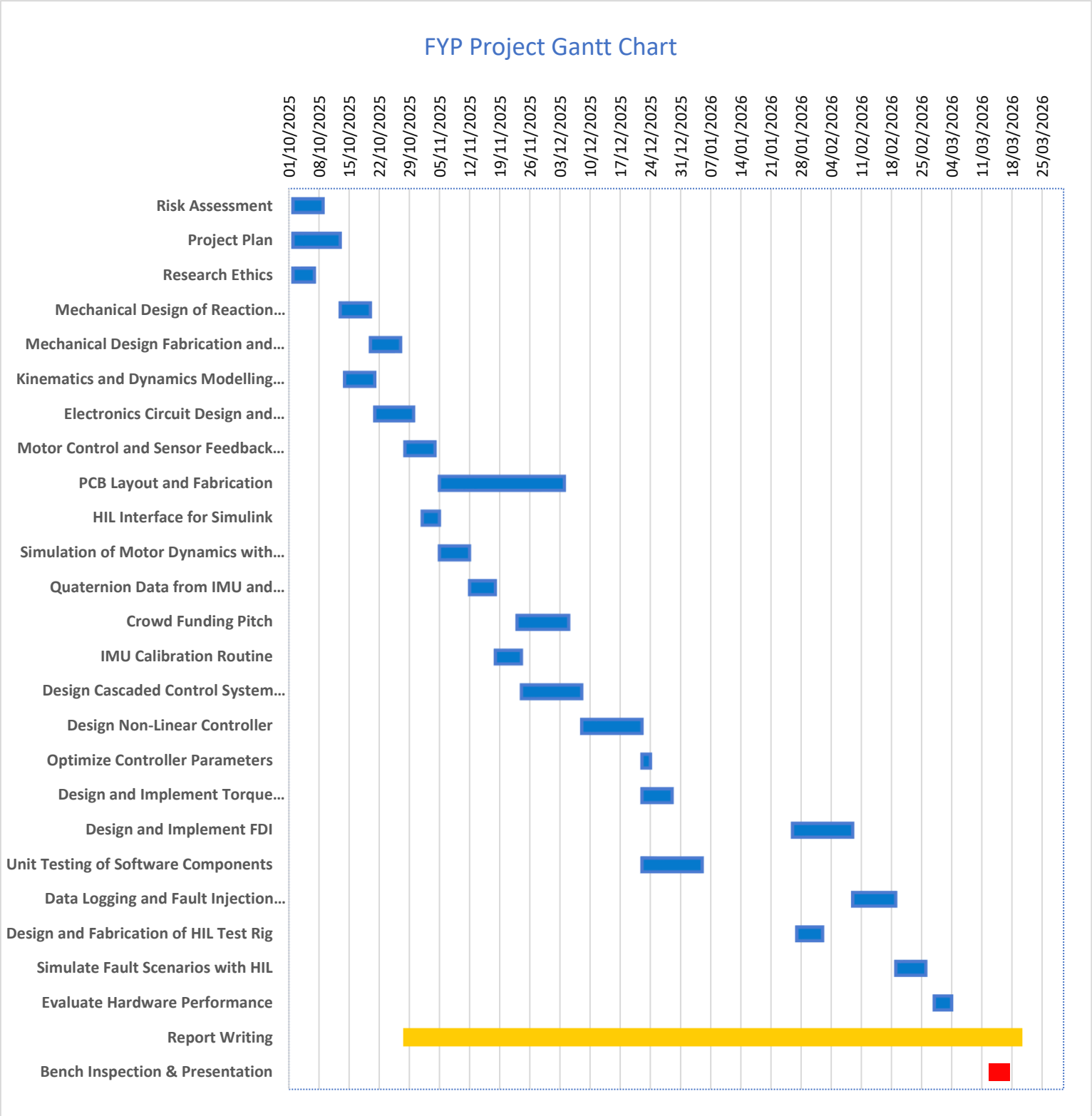
Skills Audit:

Skill	Level	Training needs
Embedded C++ programming	Moderate	Online tutorials
Dynamics and Linear Algebra	Beginner	Online tutorials, textbooks, professor available if needed
MATLAB + Simulink	Moderate	Online tutorials
Control systems	Moderate	Online tutorials, textbooks, professor available if needed
Electronics + Power	Advanced	None/ Online tutorials
CAD tools (Fusion 360)	Moderate	Online tutorials
PCB Design	Moderate	Online tutorials

Resource Audit:

Resource	Availability	Alternatives
3D Printing	Access to FDM printers in labs. Can have long lead times towards bench inspection	Laser cutting
MATLAB and Simulink	University has student licenses for MATLAB + SIMULINK	C++, Python with public libraries (OpenGL for example)
Microcontroller + various electronic components (hobby spec/COTS)	Widely available through department.	Other parts can be used to varying degrees of success
Basic lab equipment (Oscilloscope + bench power supply)	Available in fourth floor labs	Other areas of the EEE have similar equipment (3rd floor labs, LASER lab)

Appendix 1. A Gantt chart (1 Page)



Appendix 2. The risk assessment form.

Risk Assessment.pdf

Appendix 3. FYP ethics self-assessment.

Not required for this Final Year Project

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